TABLE OF CONTENT

CHAPTER NO.	TITLE	PAGE NO.
	ACKNOWLEDGEMENT	iv
	ABSTRACT	v
	TABLE OF CONTENT	vi
	LIST OF FIGURES	viii
	LIST OF TABLES	viii
	ABBREVIATIONS AND NOMENCLATURE	ix
1	**INTRODUCTION**	**1**
1.1	BACKGROUND OF HIGH-CONCURRENCY TICKETING	1
1.2	THE SCALPING PROBLEM & AUTOMATED BOT EXPLOITATIONS	2
1.3	ECONOMIC DYNAMICS & PRIMARY MARKET INEFFICIENCIES	2
1.4	MOTIVATION	3
1.5	OBJECTIVES OF THE PROPOSED WORK	3
1.6	SCOPE AND DELIMITATIONS	4
1.7	CONTRIBUTIONS OF THE PROJECT	4
2	**LITERATURE SURVEY**	**5**
2.1	HIGH-CONCURRENCY E-TICKETING ARCHITECTURES	5
2.2	SCALPER BOT ECOSYSTEMS & BEHAVIORAL DETECTION	5
2.3	RATE LIMITING & CRYPTOGRAPHIC CHALLENGE SCHEMES	6
2.4	VIRTUAL WAITING ROOMS & PRIORITY QUEUE SCHEDULING	6
2.5	DISTRIBUTED LOCKING & CONCURRENCY CONTROL PROTOCOLS	7
2.6	COMPARATIVE LITERATURE ANALYSIS & IDENTIFIED RESEARCH GAPS	8
3	**SYSTEM ARCHITECTURE AND METHODOLOGY**	**9**
3.1	HIGH-LEVEL ARCHITECTURAL FRAMEWORK & TRANSACTIONAL WORKFLOW	9
3.2	PREDICTIVE SCALPING MITIGATION & BOT DEFENSE ARCHITECTURE	10
3.3	BEHAVIORAL RISK SCORING MODEL & SENSITIVITY ANALYSIS	11
3.4	RUNTIME SECURITY POLICY ENFORCEMENT & OPERATIONAL THRESHOLDS	11
3.5	DYNAMIC VIRTUAL QUEUE & PRIORITY ALLOCATION ALGORITHM	12
3.6	QUEUING THEORY, LITTLE'S LAW ANALYSIS & SERVICE CAPACITY	13
3.7	DEADLOCK-FREE LEXICOGRAPHICAL LOCKING PROTOCOL & PROOF	14
3.8	THREAT TELEMETRY LOGGING & FORENSIC INCIDENT AUDITING	14
3.9	MATHEMATICAL MODEL OF SLIDING-WINDOW LOGIC & EXPONENTIAL DECAY	15
3.10	DISTRIBUTED THREAT FINGERPRINTING & TLS ENTROPY ANALYSIS	16

TABLE OF CONTENT (CONTINUED)

CHAPTER NO.	TITLE	PAGE NO.
4	**SYSTEM DESIGN AND PROPOSED MODULES**	**17**
4.1	AUTHENTICATION & ROLE-BASED ACCESS CONTROL MODULE	17
4.2	ANTI-SCALP SHIELD & SECURITY TELEMETRY MODULE	17
4.3	VIRTUAL WAITING ROOM & SSE STREAMING MODULE	18
4.4	TWO-PHASE HOLD & IDEMPOTENT CHECKOUT MODULE	18
4.5	SPRING DATA JPA CONCURRENCY ISOLATION & ENTITY LIFECYCLE	19
4.6	QUARTZ SCHEDULER & RABBITMQ ASYNCHRONOUS PIPELINE WITH DLQ	19
4.7	RELATIONAL DATABASE SCHEMA & DATA DICTIONARY	19
4.8	HIGH-PERFORMANCE DATABASE INDEXING & QUERY EXECUTION PLANS	20
5	**IMPLEMENTATION AND VISUAL INTERFACES**	**21**
5.1	TECHNOLOGY STACK & EXECUTION ENVIRONMENT	21
5.2	CUSTOMER INTERFACES: AUTHENTICATION & REAL-TIME SEAT MATRIX	21
5.3	OPERATIONS CENTER: ADMIN DASHBOARD & ANTI-SCALP TELEMETRY	22
5.4	QUEUE CONSOLE & CUSTOMER EVENT CATALOG JOURNEY	23
5.5	RESTFUL API ENDPOINT SPECIFICATIONS & CONTRACTS	24
5.6	FRONTEND COMPONENT ARCHITECTURE & CUSTOM HOOKS LIFECYCLE	24
6	**RESULTS AND PERFORMANCE EVALUATION**	**25**
6.1	CONCURRENCY STRESS BENCHMARKS & RACE ELIMINATION	25
6.2	SCALPER BOT NEUTRALIZATION EFFICIENCY & ROC ANALYSIS	25
6.3	VIRTUAL QUEUE LATENCY, PERCENTILE DISTRIBUTION & STRESS PROFILES	26
6.4	SYSTEM BOTTLENECK PROFILING & INDUSTRIAL BENCHMARKING	26
6.5	ARCHITECTURAL FAILURE CASE STUDY & SAFEGUARDS MATRIX	27
7	**CONCLUSIONS AND FUTURE ENHANCEMENTS**	**28**
7.1	CONCLUSIONS	28
7.2	PRACTICAL INDUSTRIAL APPLICATIONS	28
7.3	SCOPE FOR FUTURE WORK	29
	REFERENCES	**30**
	APPENDIX I: SELECTED CODE EXCERPTS	**31**
	APPENDIX II: SUPPLEMENTARY TEST MATRIX & TELEMETRY	**32**
	APPENDIX III: CERTIFICATE OF COMPLETION	**33**
	APPENDIX IV: INDIVIDUAL WORK CONTRIBUTIONS (DEEPAK A)	**34**
	APPENDIX IV: INDIVIDUAL WORK CONTRIBUTIONS (THIRUMURUGAN K)	**35**
	APPENDIX V: PLAGIARISM VERIFICATION REPORT	**36**